

Web Technologies 1 (Frontend)

Week 1: Introduction to Web Technologies

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1 Overview

Week 1 introduces the course structure and grading, basic web-development tools, how the Internet and the World Wide Web work, the front-end/back-end split, and the very first HTML document.

2 1. Course Overview

A 10-week course covering HTML5, CSS3, Bootstrap, JavaScript and jQuery, along with website-building principles and basic web design. By the end you should be able to:

- develop, validate and debug interactive websites using HTML, CSS, Bootstrap and JavaScript;
- describe basic software-development tools and concepts: debugging, source code, validators;
- deliver short, well-supported presentations and documents;
- discuss that every problem has multiple solutions, each with trade-offs.

2.1 Grading

- **1st attestation (100%):** Assignment 1 HTML & CSS Basics (15%), Assignment 2 Advanced CSS: Flexbox & Grid (15%), Assignment 3 Responsive Web Design: Media Queries + Bootstrap Grid (15%), Quiz 1 (15%), Midterm Project (40%).
- **2nd attestation (100%):** Assignment 4 Forms & Validation (15%), Assignment 5 JavaScript Basics/DOM (15%), Assignment 6 jQuery (15%), Quiz 2 (15%), Endterm Project (40%).
- **Final exam:** Final Project, oral defense with presentation.
- **Total:** $0.3 \times \text{1st Att} + 0.3 \times \text{2nd Att} + 0.4 \times \text{Final}$.
- Attendance minimum 70%; late work scores 0; automatic fail if Midterm/Endterm < 25 ; cheating/plagiarism scores 0.

3 2. Web Development Tools

- **Visual Studio Code** — main code editor, strong extension support and built-in Git integration (Sublime Text is a lighter alternative).
- **Browser Developer Tools** — inspect HTML/CSS, read console errors, test responsive behavior across screen sizes.
- **Git and GitHub** — track project history, collaborate, and publish for free with GitHub Pages.

4 3. Internet and WWW

- **Internet** — a global network of computers.
- **World Wide Web (WWW)** — the collection of websites reachable over the Internet.
- **Client–Server model:**
 - *Client* (browser) sends a request (e.g., HTTP) to a server.
 - *Server* responds with the web page.
 - In between, data is split into packets that travel through ISPs and multiple routers before reaching the destination web server.

5 4. Introduction to the Frontend

- **Front-end** — what the user sees; built with HTML and CSS.
- **Back-end** — how the site works behind the scenes, on the server and database.
- The browser renders whatever it receives:
 - **HTML** determines how the page is *structured*.
 - **CSS** controls presentation (how it *looks*).
 - **JavaScript** controls behavior (how it *functions*).

6 5. What is HTML?

HTML (HyperText Markup Language) is the standard markup language for building web pages.

- HTML defines the structure and layout of a page's content.
- HTML consists of a series of *elements*, represented by *tags*.
- Tags label content — `<h1>` marks a heading, `<p>` a paragraph, `<a>` a link, etc.
- Browsers do not display the tags themselves; they use them to render the page.

6.1 HTML Elements and Tags

An element is defined by a start tag, some content, and an end tag:

```
<tagname> Content goes here... </tagname>  
<h1> My First Heading </h1>
```

Tags are written inside < > and tell the browser what kind of content follows.

6.2 HTML Attributes

All HTML elements can have attributes, which provide additional information and are always specified in the start tag, usually as name/value pairs:

```
<tagname attribute="value"> Content </tagname>  
<a href="example.com"> Visit </a>
```

7 6. The Basic HTML Structure

An HTML document is built from three container elements:

```
<!DOCTYPE html>  
<html lang="en">  
  <head>  
    <meta charset="UTF-8">  
    <title>My First Webpage</title>  
  </head>  
  <body>  
    <h1>Hello, AITU!</h1>  
  </body>  
</html>
```

- <html> — the main container; everything belonging to the page goes inside it.
- <head> — information for the browser (title, metadata, settings); not visible on the page.
- <body> — the content actually displayed: text, images, links, and other elements.

8 7. Headings and Paragraphs

```
<h1>Page title</h1>  
<h2>Section titles</h2>  
<h3>Sub-sections</h3>  
<h4>Heading 4</h4>  
<h5>Heading 5</h5>
```

```
<h6>Heading 6</h6>  
<p>Paragraph text always starts on a new line.</p>
```

Headings (<h1>–<h6>) are titles/subtitles shown on the page, from most to least prominent. Paragraphs (<p>) always start on a new line and hold a block of text.

9 8. Breaks and Horizontal Rules

```
<p>The 1st line <br> The 2nd line</p>  
<br>  
<p>The 3rd line</p>
```

-
 inserts a single line break.
- <hr> defines a thematic change in the content (a horizontal rule).
- Neither tag has a closing tag.

10 Summary

Week 1 covers course logistics, essential tools (VS Code, DevTools, Git/GitHub), how the Internet/WWW/client-server model work, the HTML/CSS/JS division of responsibilities, and the basics of HTML: elements, tags, attributes, the <html>/<head>/<body> skeleton, headings, paragraphs, breaks and horizontal rules.